

Amritesh Sahu

Data Science Undergraduate — AI/ML + Backend Systems — LeetCode Enthusiast

Bengaluru, India — amriteshsahu96@gmail.com — +91-8252406988
GitHub — LinkedIn

Education

B.E. in Data Science, RV College of Engineering, Bengaluru
Class XII (CBSE), Delhi Public School, Ranchi

2023 – Present
2022

Projects

Gemma UI — Local LLM Interaction Interface (Ollama + Gemma 4)

- Built a fully local LLM interface using React and Vite integrated with Gemma 4 via Ollama (no cloud dependency)
- Implemented streaming responses using fetch and ReadableStream for real-time token-level output rendering
- Designed document ingestion pipeline (PDF, DOCX, PPTX) using Node.js to enable RAG-style querying over user files
- Integrated D2 (WebAssembly) to generate and render system diagrams from model output
- Resolved React state race conditions during streaming using refs and controlled updates
- Developed local persistence with conversation history, search, and editing using browser storage

Multi-Agent Medical Assistant — AI-Powered Healthcare System

- Developed a multi-agent medical assistant using LangGraph and FastAPI with intelligent routing across RAG, web search, and medical imaging agents
- Built an advanced hybrid RAG pipeline with Qdrant vector database, semantic chunking, BM25 retrieval, and cross-encoder re-ranking for medical literature search
- Integrated deep learning models for chest X-ray disease detection, skin lesion classification, and brain MRI tumor analysis using PyTorch
- Implemented voice-enabled interaction with speech-to-text and ElevenLabs text-to-speech integration for accessible medical assistance
- Designed a human-in-the-loop validation workflow and safety guardrails system for secure and reliable AI-assisted medical analysis
- Deployed the system using Docker and FastAPI with support for real-time image uploads, session management, and scalable API endpoints

Woodcraft Furniture Store — Full-Stack E-Commerce & ERP Platform

- Built full-stack platform using Next.js, MongoDB, and Redis combining storefront with internal ERP
- Designed order pipeline covering cart, checkout, inventory updates, and automated invoice generation
- Implemented role-based authentication (NextAuth.js) and Redis-based rate limiting for API protection
- Developed admin dashboards for sales, orders, and analytics using Recharts
- Engineered inventory tracking system with audit logs for consistency across operations

Achievements

- Strong foundation in Data Structures and Algorithms
- Active contributor to technical projects and hackathons
- Awarded Rotaractor of the Year (2024-2025)

Skills

Languages: Java, Python, JavaScript

Frontend: React, Next.js, Vite, HTML, CSS

Backend: Node.js, Express, FastAPI, Flask

Databases & Storage: MongoDB, MySQL, Redis, localStorage

AI/ML: Scikit-learn, TensorFlow Lite, CNNs, LSTM, Transfer Learning, RAG

Systems & Tools: Ollama, REST APIs, Streaming (ReadableStream), Git, Vercel, Firebase, Google Earth Engine

Leadership

International Services Director , Rotaract Club of RVCE	2025 - 2026
• Led cross-club collaborations and managed international partnerships • Coordinated events and strengthened community engagement initiatives	